

NISHANTH SUNDARAN

AI & Robotics Engineer | Humanoid Systems, VLA & ROS 2 / MoveIt 2

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🇮🇳 Indian (work permit for Germany: § 20 AufenthaltG, Post-Study)



PROFILE

M.Eng. AI graduate specialising in collaborative robotics, real-time control, and deep learning. Master's thesis (THD): ROS 2 / MoveIt 2 stack on a UR3e cobot with hybrid admittance control, ISO/TS 15066 safety, YOLOv8 perception, and a 13-state FSM for assistive manipulation. Recent portfolio: Vision-Language-Action (VLA) architectures, cross-embodiment transfer, and human-to-robot motion retargeting on Unitree G1 and H1. Additional depth in SLAM/ROS mobile robotics, OpenCV computer vision, and IoT sensor fusion. Seeking a full-time role in robotics or AI engineering in German industry.

WORK EXPERIENCE

- 09/2025 – 06/2026 Technische Hochschule Deggendorf (THD)**, Technologie Campus Cham, Germany
Master's Thesis: Human-Robot Interaction with Force Feedback (Supervisor: Prof. Dr. Dmitrii Dobriborsci; defended 14.06.2026)
- Real-time hybrid position-admittance controller at 500 Hz for safe HRI on a UR3e cobot in ROS 2 / MoveIt 2.
 - ISO/TS 15066-compliant safety: speed and separation monitoring (SSM), power and force limiting (PFL), Hand Guiding.
 - RealSense D435i vision pipeline: YOLOv8 human detection, hand-eye calibration, HSV object detection with depth back-projection.
 - 13-state FSM for assistive lift-and-place with autonomous grasping and force-guided placement.
- 12/2023 – 11/2024 Energy Future World**, London, United Kingdom (Remote)
Machine Learning Intern
- Aerial-imagery CV for renewable-energy infrastructure tracking; 85% detection accuracy with OpenCV, Scikit-learn, NumPy, Pandas.
 - Image-processing pipeline (edge detection, contrast enhancement) and ML-model benchmarking on real-world datasets.
- 03/2022 – 12/2022 Amazon**, Bangalore, India
Process Associate
- Data analysis and auditing on operational trends; contributed to automation and forecasting workflows.
- 05/2019 – 11/2021 Power And Instrumentations Ltd**, Gujarat, India
Project Engineer
- Analysed HV power-grid performance for reliability; managed cost, quality, and resources across project sites.

EDUCATION

- 10/2022 – 06/2026 M.Eng. Artificial Intelligence for Smart Sensors and Actuators**, Technische Hochschule Deggendorf. Focus: ML, Deep Learning, Computer Vision, Autonomous Systems, IoT, Robotics. Thesis: *Human-Robot Interaction with Force Feedback* (defended 14.06.2026).
- B.Eng. Electrical and Electronics Engineering**, University of Calicut, India. Focus: DSP, Digital System Design, Power Electronics, IoT.

TECHNICAL SKILLS

Programming	Python, C/C++, Git, Linux (Ubuntu), Jupyter Notebooks, VS Code
Robotics & Control	ROS 2 (Humble), MoveIt 2, MoveIt Servo, Classical Control (PID, Admittance, Impedance), Motion Planning (OMPL, Pilz), SLAM, URDF/Xacro, tf2
Simulation	Gazebo, ROS 2 Sim, Real-hardware integration (UR3e, Robotiq 2F-140)
AI / ML	Vision-Language-Action (VLA) models, Vision-Language Models (VLM), Foundation models, Cross-embodiment transfer, Imitation learning, Representation learning, Self-supervised learning, Reinforcement Learning (PPO), PyTorch, JAX / MJX, TensorFlow, Scikit-learn, Deep Learning, Time Series (ARIMA, SARIMA)

Vision / Sensors	OpenCV, YOLOv8, Intel RealSense D435i, Depth Estimation, HSV Segmentation, Hand-Eye Calibration, Point Clouds
Safety	ISO/TS 15066 (SSM, PFL, Hand Guiding), Risk Assessment, Functional Safety
SW Engineering	Git version control, Code review, CI, Unit testing, Documentation, Publication-quality writing
Data	NumPy, Pandas, Matplotlib, Feature Engineering, Predictive Modelling
Soft Skills	Analytical Thinking, Problem Solving, Self-driven & Quick Learner, Cross-functional Collaboration, Technical Communication, Agile & Scrum, Adaptable to Fast-paced Environments, Attention to Detail

LANGUAGES

English	Professional fluency (C1)
German	A2 (elementary), actively learning toward B1

CERTIFICATIONS

08/2026	AI Agents and Agentic AI with Python & Generative AI, Vanderbilt University. <i>Skills:</i> Agentic AI, Tool Use, Function Calling, AI Agent Architecture, LLM Application Development, Prompt Engineering, Prompt Patterns, OpenAI API, Retrieval-Augmented Generation (RAG), Python for AI. Verify credential.
09/2026	Machine Learning with Python, IBM. <i>Skills:</i> Supervised Learning, Unsupervised Learning, Classification (Logistic Regression, KNN, SVM, Decision Trees), Regression (Linear, Non-linear), Clustering (k-means, Hierarchical, DBSCAN), Recommender Systems, Model Evaluation (Cross-validation, Precision/Recall/F1), Scikit-learn, Pandas. Verify credential.

PROJECTS

GitHub: github.com/NishanthSundaran

Human-Robot Interaction with Force Feedback

Master's Thesis | repo

- Complete ROS 2 / MoveIt 2 framework for safe HRI on a UR3e cobot with admittance control, YOLOv8 human exclusion, RealSense D435i perception, 13-state task FSM. Demo video included; supports real-hardware and Gazebo Ignition simulation launch paths.

Embodied Agent: VLM-driven Pick-and-Place

Multimodal AI | repo

- Multimodal foundation-model system: VLM grounds objects in live RGB frames, 8-tool agentic loop translates natural-language intent into action plans, MoveIt 2 executes manipulation on a UR3e with Robotiq gripper.
- Built with the Anthropic Messages API; 3D localisation via depth back-projection; real-time multimodal perception, task planning, and robot manipulation.

One Brain, Many Bodies: Humanoid Autonomy Stack in MuJoCo

Portfolio Project | repo

- Hierarchical stack: local open-source VLM (Qwen2.5-VL-3B, no cloud APIs) plans at ~ 1 Hz via tool-calling with Set-of-Mark visual grounding, compliant whole-body controller executes at ~ 500 Hz.
- Cross-embodiment transfer:** same brain drives Unitree G1 and H1 through one tool interface; only per-robot config changes. 22-DoF five-finger hands composed onto H1 at load time via MjSpec.
- Trained PPO stance-balance policy (3,300 env-steps/s, 14 CPU workers) holding pelvis to 1.4 cm while arms reach. "Human-like motion" measurable against Flash & Hogan minimum-jerk; 3/4 grasp motions meet every human-reaching criterion.

G1 Motion Imitation: Learning Human Gait from Motion Capture

Portfolio Project | repo

- Human-to-robot motion retargeting** via DeepMimic-style imitation learning on Unitree G1, tracking LAFAN1 mocap in JAX / MJX with PPO; 6 GB laptop GPU at $\sim 30,000$ env-steps/s ($9\times$ CPU baseline), 5.8B steps across 13 cycles.
- Gait within 2 to 7% of human reference on speed (0.48 vs 0.49 m/s), cadence, stride, pelvis bounce.
- PD position control reached in 5M steps what torque-control recipe needed 300M for ($\sim 60\times$ sample efficiency); contributed four undocumented framework fixes upstream, including a silent JAX-scalar vs MuJoCo-enum bug blocking all dataset loads.

AQI Prediction: Time-Series ML Pipeline

Case Study | repo

- ML pipeline for multi-city air-quality forecasting with ARIMA and SARIMA; over 93% test accuracy. Seasonality decomposition, cross-dataset validation, reproducible Jupyter workflow.

Autonomous Mobile Robot with SLAM Navigation

Case Study | repo

- Four-wheeled LiDAR robot on ROS 2 Humble: slam_toolbox online SLAM and a full Nav2 stack (AMCL, NavFn, DWB); 3/3 goals reached with 0 collisions in a cluttered warehouse simulation.

Additional Projects

Self Study

- Histogram Equalization:** classical image-processing for contrast enhancement under varied lighting. **Object Detection with YOLO + Audio Alert:** YOLOv8 + gTTS with a live-annotated GUI. **IoT Speed-Detection:** accelerometer + GPS sensor box streamed to ThingSpeak.